Soohyun Choi

Integrated M.S./Ph.D. Student in Electronic Engineering | Reinforcement Learning

Hanyang University | Information and Intelligence Systems Lab (IISL) | Seoul, Republic of Korea

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Research Interests

My research interests lie in reinforcement learning, particularly in the mathematical and geometric analysis of learning algorithms. I am also interested in developing new algorithms and modeling frameworks using generative models and stochastic processes, especially for goal-conditioned and long-horizon control.

Education

Hanyang University Mar. 2024-Present

Integrated M.S./Ph.D. Program in Electronic Engineering

Information and Intelligence Systems Lab (IISL) | Advisor: Prof. Songnam Hong

Hanyang University 2018-2024

B.S. in Electronic Engineering

Manuscripts & Research Projects

PathBridger: Subgoal Bridges for Offline Goal-Conditioned Reinforcement Learning 2026

Soohyun Choi, Seonvin Cho*, and Songnam Hong | Under review | *Equal contribution | Code*

Developed a hierarchical offline GCRL framework that connects subgoal selection to short-horizon execution through state-space bridges, inverse dynamics, and execution-time replanning.

ALARM: Attention and Line Search Based Analog Circuit Optimization on a Riemannian Manifold 2024; rev. 2026

Soohyun Choi, Minseok Oh, and Ickhyun Song | Undergraduate capstone technical report | Report

Formulated designer-selected performance equivalence classes and quotient geometry for circuit search; achieved 10/10 success with 33.2 SPICE evaluations on average in the archived VCO study.

Ongoing Research

Multi-step Proximal Policy Improvement 2026-Present

Study re-centered, critic-guided proximal refinement and how step size and refinement depth shape stability across policy-improvement budgets and critic regimes. [Code]

PathFlower: Goal-Conditioned State Flows Ongoing

Extend PathBridger toward final-goal-conditioned generative state flows that produce useful intermediate states for long-horizon offline control.

Adaptive Policy Optimization Ongoing

Develop adaptive policy refinement that determines when and how far to update under uncertain critic guidance.

Honors & Funding

Top Excellence Award - Alumni Association Special Award (KRW 1,000,000) 2023

Top Excellence Award - 15th Capstone Design Fair (KRW 1,000,000) 2023

Graduation Honors, Hanyang University 2024

BK Research Assistantship, Hanyang University Fall 2025

Selected Graduate Coursework

Learning & Control: Machine Learning; Robot Learning; Advanced Optimal Control Theory

Probability & Analysis: Probability and Statistics; Stochastic Processes; Uncertainty Quantification; Functional Analysis I